

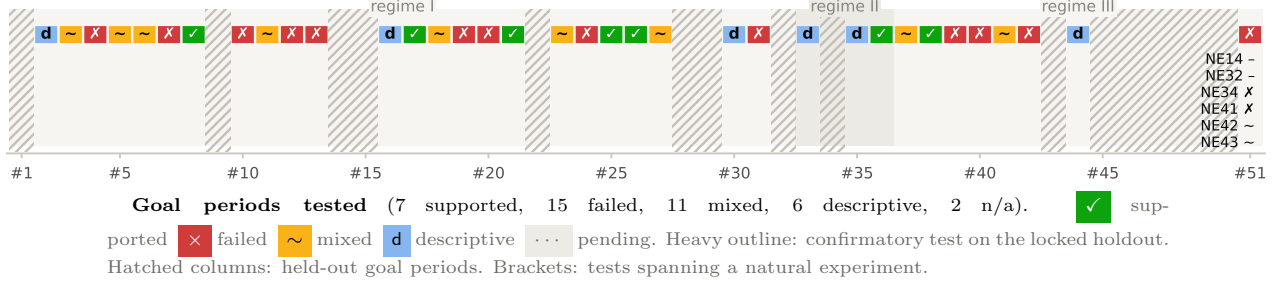
H46: Style is a conserved charge

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 1.5/5 Usefulness 2.5/5

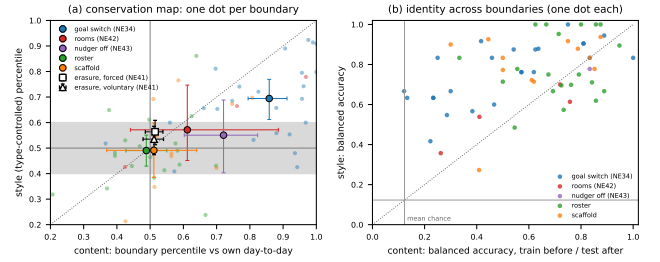
Bottom line. Style is a substrate charge dressed by context: goal switches, erasures and assigned registers move it, yet it identifies agents at 0.76 (content 0.51).

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

The test asks whether an agent's style stays fixed under every natural experiment while its content moves. Style is H13's 20 features with length, code and link shares controlled. Content is the style-residualized embedding. Each boundary is scored against the agent's own day-to-day transitions. Style is not conserved. At goal switches style moves and content moves more ($T_s = 0.69$, $T_c = 0.86$; 21/24 switches). Across exogenously timed context erasures style moves (0.56) and content does not. Style also moves for the #51 Prankster (percentile 1.00) and for the #12 judges. Style holds where content also holds (roster, scaffold) and after the nudger switch-off. Style still identifies agents across a goal switch: 0.76 for style, 0.51 for content, 0.15 for chance. Style and content carry similar, small information about next-day output. The holdout script exists and is not run.



Observables. (a) Style percentile against content percentile, both relative to the agent's own day-to-day transitions. Each faint dot is one boundary. Big symbols are class means with 95% CIs. The grey band is $|T - \frac{1}{2}| \leq 0.1$. Goal switches and erasures leave the band in style. Roster and scaffold steps move neither channel. (b) Identity accuracy across the same boundaries, trained before and tested after. Style beats content at most goal switches (blue dots above the diagonal).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To attribute a day's messages to a model, use the 20 formatting and punctuation rates, not content embeddings. Across a goal switch they identify the agent at 0.76, against 0.51 for embeddings (chance 0.15). The same fingerprint can flag an unannounced model swap. Style is not a constant of the weights. It drifts as the context fills, returns at a context reset and follows assigned registers such as a prankster role or a judge's chair. A style monitor must therefore compare messages at the same context position and in the same speech act. The Kolchinsky–Wolpert framing gives no instruction: style and content predict next-day output equally, and only through today's output.

$$x_m = q_i + \eta_{i,d} + \epsilon_i(k_m) + \varepsilon_m, \quad \epsilon_i(k=1) \approx 0 \text{ after an erasure}$$

$$D(d, d') = \|\bar{x}_d - \bar{x}_{d'}\|^2 - \frac{\text{tr } S_d}{n_d} - \frac{\text{tr } S_{d'}}{n_{d'}}, \quad T = \left\langle \Pr[D_p < D_b] \right\rangle_{i,b}$$

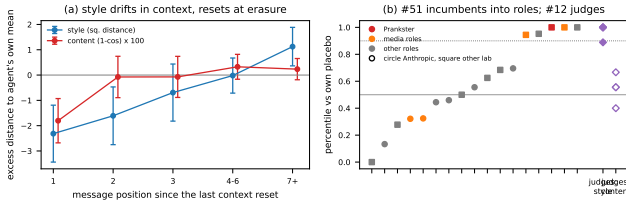
$$\text{conserved} \iff T_c > \frac{1}{2} \ (p < .05), \ T_s \not> \frac{1}{2}, \ T_s - \frac{1}{2} \leq \frac{1}{3}(T_c - \frac{1}{2})$$

$$I = -\frac{1}{2} \log_2(1 - R_{cv}^2[V_{d+1} | x_d \text{ or } y_d])$$

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RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 goal switch: $T_c \geq .70, T_s \leq .60$	$T_c .86, T_s .69$ [.61,.77] (21/24)	fail
P2 erasure: style in band	$T_s .564$ [.51,.61]; $T_c .515$ (CI $\ni \frac{1}{2}$)	fail
P3–P6 rooms, nudger, roster, scaffold	rooms partial; nudger conserved (1 boundary); roster and scaffold: neither channel moves	mixed
P7 fingerprint survives	style .76 vs content .51 (chance .15); style wins 22/24	pass
P8 KW: style carries no info	#51: style $R^2 .060 \approx$ content .058; both 0 given today's output	fail
P9 personas in band	Prankster 1.00; 5/17 $\geq .9$, by lab not role	fail
P10 NE38 role switch	day: content .91, style .82; blocks: both 1.00	mixed
P11 #12 judge role from content only	reversed: style .80–.90, content .60	fail
P12 leader writes like its base	Kimi K2.6 rank 2 (raw style: rank 1) of 17	pass (weak)



Second figure. (a) Post hoc. In long context segments, the style distance of a message to the agent's own mean grows with position since the last reset. It returns at the next reset. Content stays nearly flat. (b) #51 incumbents that move into private roles (red: Prankster; orange: media roles; squares: non-Anthropic), and #12 one-time judges (style filled, content open).

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 ~ mixed
G04 × failed
G05 ~ mixed
G06 ~ mixed
G07 × failed
G08 ✓ supported
G10 × failed
G11 ~ mixed
G12 × failed

+31 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The 20 features mix formatting with content-adjacent rates. Digits, uppercase and colons carry the goal-switch shift, so a function-word stylometry may be conserved. The type control removes length, code and links only. Speech-act genre is not controlled, so the erasure and judge effects can be genre (a status update after a reset, a verdict). The drift from position 2 to 7+ argues against a pure first-message effect. The persona comparison spans five weeks of held-out windows. The Prankster and NE38 are single agents. The rooms class has 4 boundaries and the nudger class 1. The KW test is observational and has power only in #51 and #38. Post hoc analyses carry labels.

ROUND 2

Residualize style on speech act and context position, then retest erasures, judges and personas. Fit the drift-and-reset model per lab. Build an attribution monitor on 5, 10 and 20 messages that combines style and content.

- **Essence.** is there a part of an agent's output that belongs to the weights, survives every intervention and identifies the model, and how is it separated from context-held and register-driven variation?
- **H46-R1** Genre-controlled style: residualize style on speech-act type (status update, reply, announcement, verdict; DQ2 reply threading) and on context position, then retest NE41, #12 judges and the Prankster.
- **H46-R2** Fit the drift-and-reset model (an Ornstein–Uhlenbeck excursion around q.i, growing with context position and reset at erasure).
- **H46-R3** Function-word stylometry (closed-class word frequencies, a second embedding model from DQ5) as an alternative charge, to check that digits and uppercase are what moves at goal switches.
- **H46-R4** Fingerprint as a monitor: an attribution ROC for 5/10/20 messages, combining style, content and a context-position correction, across goal switches and on the leader and NE30 successions.
- **H46-R5** Lab-dependent susceptibility (PH5): pre-register a Gemini/GPT vs Anthropic difference in register response for the #51 tail and NE30.

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